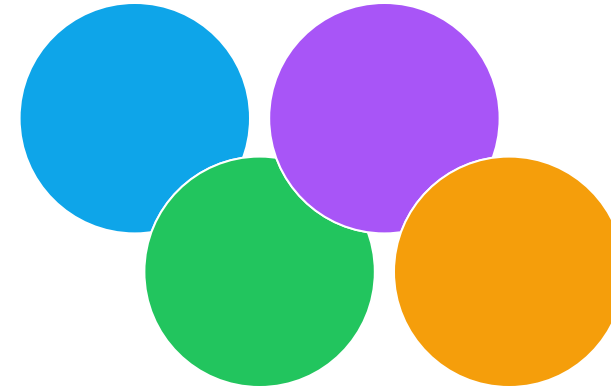


⁰¹ NetGuard AI

Predictive Network Failure and Anomaly
Monitoring

Achmad Maulana - 241730016



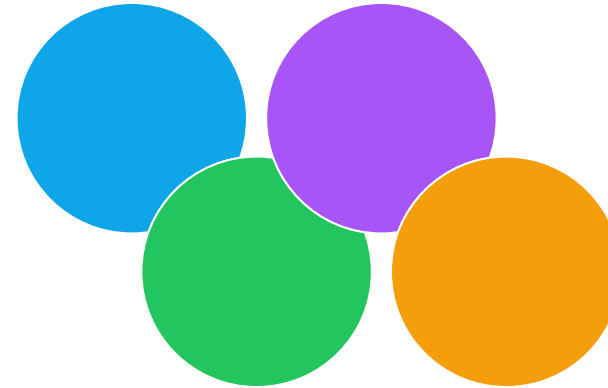
AI

Network
Security

CICIDS2017

02 Masalah

Monitoring jaringan kampus/lab sering masih manual, sedangkan trafik abnormal dapat muncul cepat dan sulit dibaca dari log mentah.



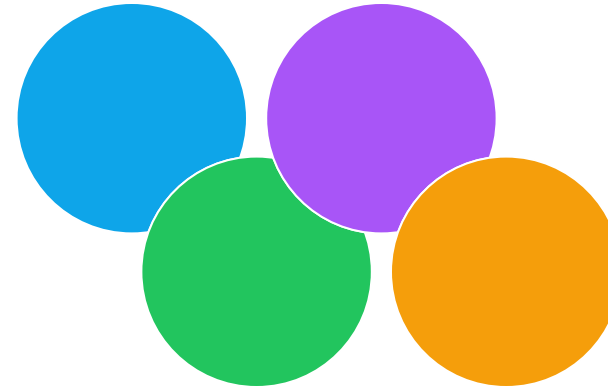
Low Budget

Praktis

Reproducible

03 **Research Gap**

Banyak penelitian IDS kuat pada metrik, tetapi belum selalu menyediakan dashboard edukatif, risk score, rekomendasi tindakan, dan paket artefak yang mudah direplikasi mahasiswa.

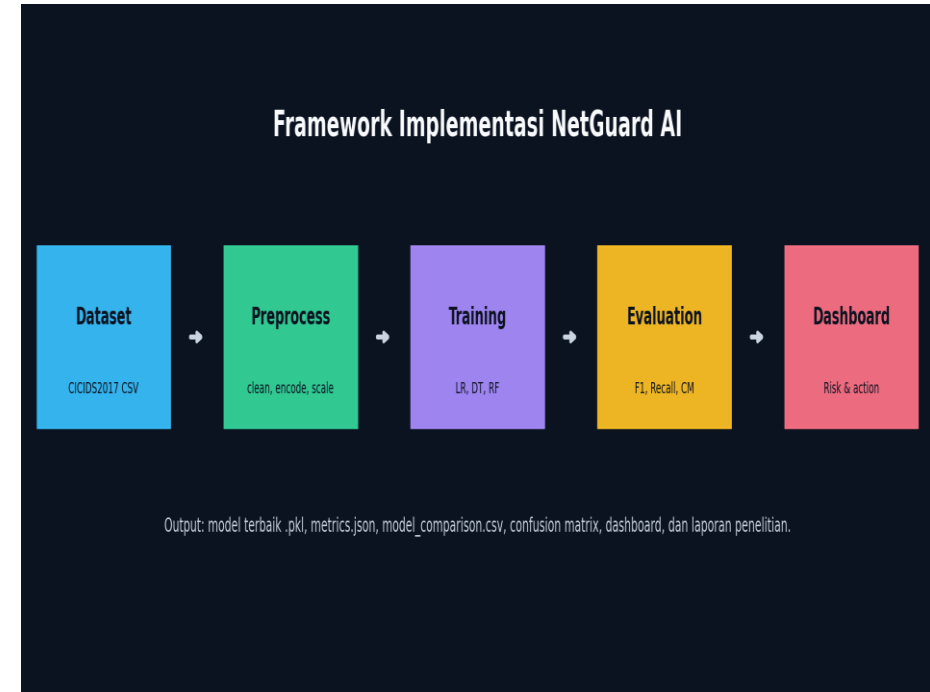


Gap

Novelty

04 Novelty

Integrasi deteksi anomali machine learning dengan dashboard monitoring ringan, klasifikasi risiko otomatis, dan rekomendasi tindakan untuk administrator pemula.



Dashboard

Risk Action

05 Dataset

CICIDS2017 Friday Afternoon DDoS subset
representatif.

Processed records: 19.935

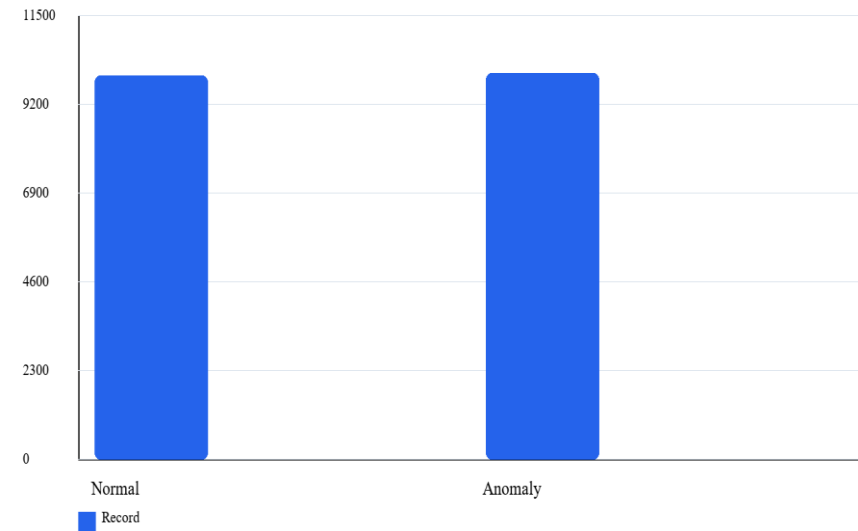
Normal: 9.935

Anomaly: 10.000

Public Dataset

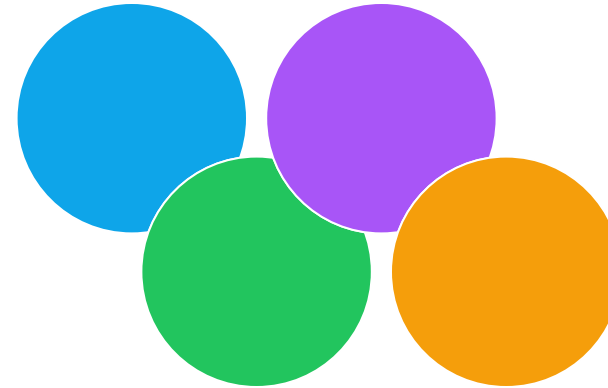
Binary Class

Distribusi Dataset CICIDS2017 Subset



⁰⁶ Preprocessing

Normalize columns -> label encoding ->
numeric features -> handle infinity/missing
values -> median imputation -> remove
duplicates -> export processed CSV.



Pipeline

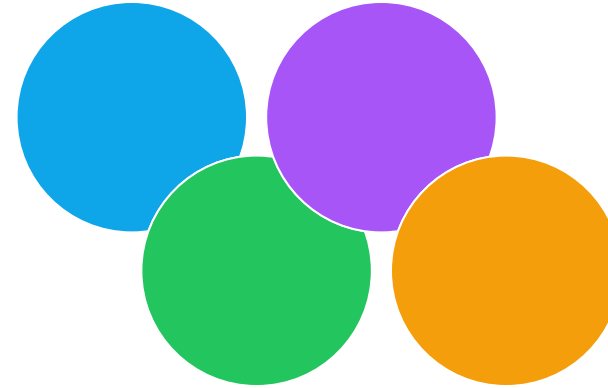
07 **Model**

Baseline: Logistic Regression

Proposed practical model: Decision Tree /

Random Forest comparison

Selection: F1-score, Recall, Accuracy



LR

DT

RF

08 Hasil Utama

Best model: Decision Tree

Accuracy: 0.9992

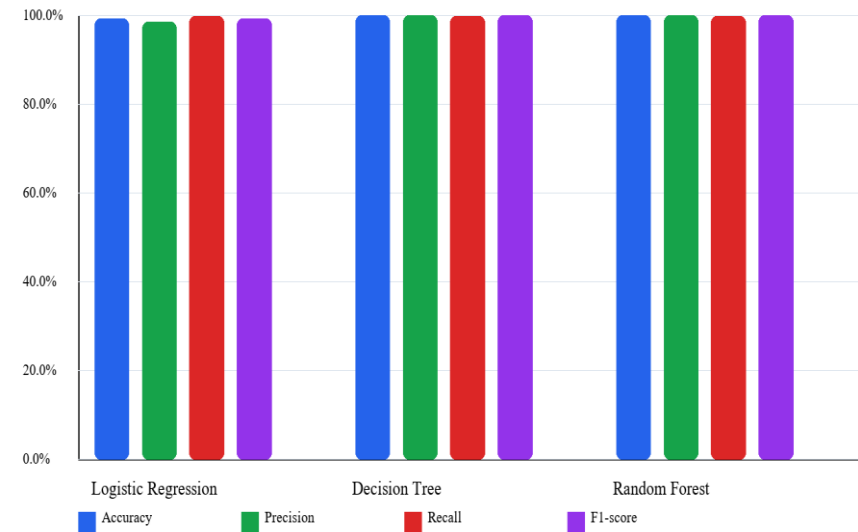
Precision: 0.9995

Recall: 0.9990

F1-score: 0.9992

Best Model

Perbandingan Model NetGuard AI



09 Confusion Matrix

Kesalahan klasifikasi sangat rendah pada data uji. Fokus evaluasi tetap pada false negative karena anomaly yang lolos deteksi berisiko terhadap layanan.

Evaluation

Confusion Matrix - Decision Tree

	Pred Normal	Pred Anomaly
Actual Normal	1986	1
Actual Anomaly	2	1998

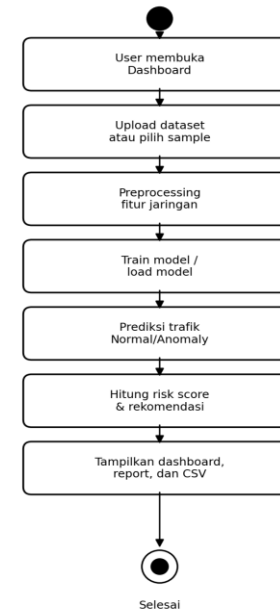
Interpretasi: kesalahan klasifikasi sangat rendah pada data uji subset CICIDS2017.

10 Dashboard

Dashboard menampilkan total record, normal traffic, anomaly traffic, model accuracy, risk score, risk level, recommended action, comparison chart, dan confusion matrix.

Flask

Chart.js



¹¹ Risk Score

Low: anomaly ratio $< 10\%$

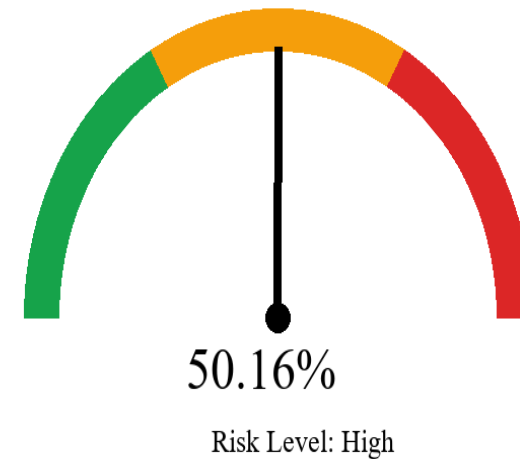
Medium: 10% sampai $< 30\%$

High: $\geq 30\%$

Pada batch eksperimen, risk level masuk High.

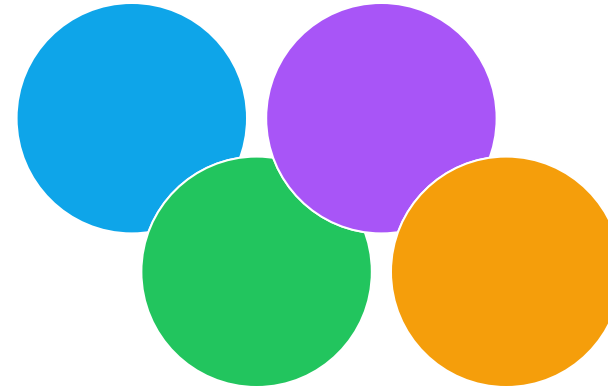
Operational

Risk Score Based on Anomaly Ratio



¹² Deployment

Mode sederhana: Flask lokal untuk demo dan VPS/Render/Railway untuk akses online. Domain dapat diarahkan melalui Cloudflare DNS ke server agar laptop tidak perlu menyala.



Domain

Server

13 Keterbatasan

Masih memakai subset DDoS, belum cross-dataset, belum tuning sistematis, dan belum diuji pada trafik lokal kampus dengan izin etik.

Limitation

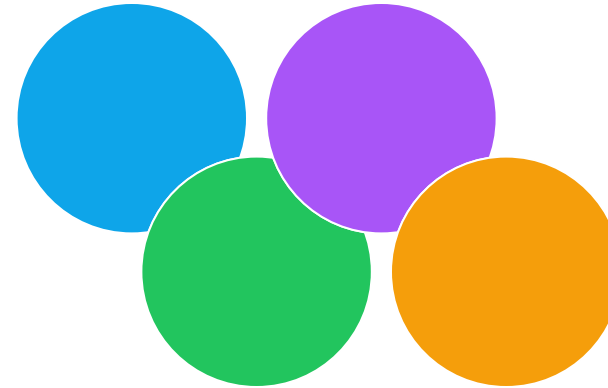
Top Feature Association with Label



Catatan: skor ini berbasis korelasi absolut pada subset terproses; gunakan feature importance model untuk analisis lanjutan.

14 **Penutup**

NetGuard AI memenuhi siklus penelitian: literature mapping, gap, novelty, implementasi AI, eksperimen, dashboard, artefak, dan draft IEEE.



**Ready for
Defense**